

Yucheng BAO

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Education

The Hong Kong Polytechnic University

Sep 2021 - Jun 2025

Qualification: BSc (HONS) Scheme in Computing

Overall Weighted GPA: 3.62/4.3 with First Class Honors

Dean's Honor List of Faculty of Engineering

Minor in Applied Mathematics

The Chinese University of Hong Kong

Sep 2025 - Jun 2026

Qualification: MSc in Computer science

Research Experiences

LLM inference optimization: KV Cache Compression

Hong Kong, Nov. 2024 - Apr. 2025

Technologies: vLLM, cuSZp

Background: Prof. Zhaorui ZHANG and her research team aim to find a efficient method in optimizing LLM inference without sacrificing output quality too much. KV Cache Compression is an important part of this work, and becomes my final year project. This project was also under the instruction of Dr. Xin YAO from Huawei.

- Tested cuSZp algorithm step by step(including quantization&prediction,fixed-length encoding, global synchronization , and Block Bit-shuffle) and evaluated its compression rate & error bound;
- Used vLLM to host LLM, and went through its python interfaces;
- Modified the interfaces of flash attention and applying cuSZp without modifying vLLM kernels. When prefill phase completed, compress and store the compressed KV Cache. When decode phase starts, compression and decompression of KV took turns repeatedly;
- Tested and evaluated the perplexity of output on wikitext-103, and compression ratio vs error bound

Outcome: The KV cache has been compressed to 40% of its original size, and the output quality was guaranteed;

Retrieval-Augmented Generation (RAG)

Hong Kong, May 2024 - Sept. 2024

Position: Research Assistant in the Faculty of Engineering, PolyU

Technologies: pdf plumber, OCR, DocX Chain, Embedding Models, Vector Databases, Keyword Search

Background: Prof. Dan WANG and his research team aim to develop a system for predicting the embodied carbon of manufacturing devices (CPU, Memory, Storage) and building an advanced question-answering (QA) system;

- Extracted and processed data from PDF documents using tools such as pdf plumber, Optical Character Recognition (OCR), and open-source solutions like Alibaba's DocX Chain, ensuring comprehensive text extraction from industry reports;
- Tested and evaluated various embedding models, implementing a local vector database for efficient document retrieval;
- Explored the impact of different distance functions on vector retrieval accuracy;

- Independently implemented a hybrid search mechanism combining vector retrieval with keyword search, enabling precise queries on massive text datasets;
- Designed and implemented a computational module to calculate embodied carbon values for CPU, Memory, and Storage components;

Outcome: The RAG system accurately answers queries in industry-specific databases; Result in a two-page publication: <https://doi.org/10.1145/3671127.3698699>

Internship

Huawei Hong Kong Research Center - Theory Lab

Hong Kong, Oct 2025 - Jun. 2026

Position: Research Intern

Technologies: NPU, cuda, Operator scheduling, RPC, shared Memory

Background: Huawei is developing a FlexNPU system with the goal of intercepting CANN (Compute Architecture for Neural Networks) APIs to achieve elastic utilization of NPU resources. The system adopts a client-server architecture, intercepting and forwarding CANN API calls to achieve NPU resource sharing, state management, and elastic scheduling.

- Implementation of the shared memory transmission model, splitting the channel into a global control area, a slot metadata area, and small/large data planes, supporting fast paths for small packets and segmented allocation for large packets.
- Implemented a client thread reuse mechanism based on thread-local slots, and an execution model on the server side that binds workers to active slots, optimizing concurrent request processing efficiency and latency.
- Designed and refactored the thread architecture within the server process, using polling threads to poll status bits and worker threads to adaptively wait, which successfully alleviated the problem of excessive CPU load on the server side.

Outcome: CPU load dropped significantly from 300% to 160%, reducing resource consumption by approximately 47%. LLM inference throughput decreased by only 3.87%.

Optional Information

Noteable Awards:

- Hong Kong Government Scholarship - Talent Development Scholarship
- Dean's Honor List of Faculty of Engineering, PolyU;
- ASC 2024 Supercomputing Challenge First Prize;
- ASC 2024 Supercomputer Challenge Group Award;
- Outstanding Student of Artificial Intelligence Training Camp in 2022 Summer Exchange at Shanghai Jiaotong University;
- 2020 CphO (Chinese Physics Olympiad) Second Prize;

Programming Skills:

- Proficient in AI-related Python Programming (pytorch, langchain numpy, pandas, sklearn, skimage, etc.); familiar with vLLM & SGLang framework.
- Familiar with C++

Languages: Native in Chinese & Fluent in English with IELTS 6.5

Research Interest: Large Language Models (KV Cache Compression & Inference Optimization), Computer system